

1. Administrative & Study Identification

Full Study Title: Phase 1, 2-Cohort, Open-label, Fixed-sequence, Drug-drug Interaction Study to Evaluate the Effect of Repotrectinib on the Pharmacokinetics of Transporter and CYP P450 Probe Substrates in Healthy Adult Participants **Brief Study Title:** Study to Evaluate the Effect of Repotrectinib on the Drug Levels of Transporter and CYP P450 Probe Substrates in Healthy Adult Participants **Protocol Number:** CA127-1088 **NCT Number:** NCT07223671 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Repotrectinib **Phase of Development:** Phase 1 **Medical Monitor:** BMS Clinical Trials Contact Centre

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of repotrectinib on the drug levels of transporter and CYP P450 probe substrates in healthy adult participants. **Secondary Objectives:** To evaluate the safety, tolerability, and clinical interactions of repotrectinib when co-administered with these common probe substrates.

2.2 Background & Rationale Repotrectinib is a targeted oncology therapy (ROS1/NTRK inhibitor) utilized in advanced solid tumors and non-small cell lung cancer. To ensure safe clinical application in real-world patients who frequently have comorbidities, it is essential to map a clear drug-drug interaction (DDI) profile. This study investigates how repotrectinib affects the levels of commonly prescribed medicines –such as metformin, digoxin, rosuvastatin, bupropion, flurbiprofen, and omeprazole—to guide safe dosing limits and refine product labeling.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Self-controlled (Fixed-sequence baseline vs. co-administration) **Blinding:** Open-label **Randomization:** Non-randomized, parallel groups (2-Cohort design) **Trial Status:** Not Yet Recruiting

3.2 Planned Enrollment & Duration Target \$N\$: 32 healthy adult participants **Number of Sites:** United States **Estimated Study Start:** October 2025 (Regulatory submission/tracking start) **Estimated Primary Completion:** March 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must have a body mass index (BMI) of 18.0 to 32.0 kg/m² and should be between the ages of 18-60 years inclusive. 2. Healthy female (as assigned at birth) participants who are individuals not of childbearing potential (INOCBP) and healthy males with no clinically significant deviation from normal for the following: medical history, physical examination, vital signs, 12-lead ECG, and clinical laboratory assessment results as determined by the investigator. 3. Willing and able to adhere to the fixed-sequence dosing protocol and pharmacokinetic blood sampling schedules.

4.2 Exclusion Criteria 1. Participants must not have a significant history of clinical manifestation of any metabolic, allergic, dermatological, renal, hematological, pulmonary, cardiovascular, gastrointestinal, neurological, respiratory, endocrine, psychiatric, neoplastic, or genitourinary abnormalities/diseases as determined by the investigator or designee. 2. Other protocol-defined Inclusion/Exclusion criteria apply, including concomitant use of prohibited medications (specifically other known strong or moderate CYP450 or transporter inducers/inhibitors). 3. Pregnancy or breastfeeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sequential administration of oral repotrectinib and oral probe substrates across designated dosing periods. Probe substrates include Metformin, Digoxin (ATC Code: C01AA05; Trade Names: Digox, Lanoxin Pediatric, Digitek, Lanoxin, Lanoxicaps, Cardoxin), Rosuvastatin, Bupropion, Flurbiprofen, and Omeprazole. **Route of Administration:** Oral **Duration of Treatment:** Short-term exposure tailored for complete pharmacokinetic profiling of the substrates and repotrectinib.

5.2 Concomitant & Prohibited Therapy Permitted: Only the specified transporter and CYP P450 probe substrates evaluated in the study. **Prohibited:** Use of any non-study prescription drugs, over-the-counter medications, or herbal supplements (e.g., St. John's Wort) known to affect CYP enzyme or transporter activity.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Physical Exam, Labs, 12-lead ECG
Baseline	Day 0	Admission to Clinical Unit, Baseline Vitals, Safety Checks
Interim	Fixed-Sequence	Probe Substrates administered alone (PK blood draws); Repotrectinib run-in; Co-administration of Substrates + Repotrectinib (PK blood draws)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Cohort 1:** Area Under the Plasma Concentration-time Curve From Time Zero to Time of Last Quantifiable Concentration of AUC (0-T) of Probe Substrate With Repotrectinib (Probe substrate contains metformin, digoxin, and rosuvastatin). 2. **Cohort 1:** AUC (0-T) of Probe Substrate Without Repotrectinib (Probe substrate contains metformin, digoxin, and rosuvastatin). 3. **Cohort 1:** Area Under the Plasma Concentration-time Curve From Time Zero Extrapolated to Infinite Time (AUC(INF)) of Probe Substrate With Repotrectinib (Probe substrate contains metformin, digoxin, and rosuvastatin).

7.2 Secondary Endpoints 1. Number of Participants with Adverse Events (AEs). 2. Number of Participants with Serious Adverse Events (SAEs). 3. Number of Participants With Clinically Significant Physical Examination Findings.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous in-patient clinical observation during dosing and PK sampling periods. **Serious Adverse Events (SAEs):** Required to be reported within 24 hours to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Internal safety review managed by Bristol Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Concentration Population (participants with evaluable concentration-time profiles); Safety Set (all participants who received at least one dose of any study drug). **Sample Size Justification:** $N = 32$ participants is statistically powered to detect clinically significant differences in the AUC and $C_{\max}$ for the specific DDI probe substrates in a crossover/fixed-sequence design. **Handling of Missing Data:** Standard pharmacokinetic exclusion parameters applied for missing concentration timepoints; safety data handled via actual observed cases.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the ethical principles of the Declaration of Helsinki. **Monitoring Plan:** Continuous site monitoring of the clinical pharmacology unit to verify adherence

to strict PK sampling windows. **Audit Trail:** Maintained through validated electronic data capture (EDC) systems and documented bioanalytical sample chain-of-custody.

11. Study Identifiers & Governance

Primary ID: CA127-1088 **Registry Number:** NCT07223671 **Posting ID:** 1071291754880278969 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Study to Evaluate the Effect of Repotrectinib on the Drug Levels of Transporter and CYP P450 Probe Substrates in Healthy Adult Participants **Official Regulatory Title:** Phase 1, 2-Cohort, Open-label, Fixed-sequence, Drug-drug Interaction Study to Evaluate the Effect of Repotrectinib on the Pharmacokinetics of Transporter and CYP P450 Probe Substrates in Healthy Adult Participants

12. Clinical Framework (Pharmacology Specific)

Primary Condition: Healthy Volunteers **UMLS Preferred Concept:** Drug-Drug Interaction **Diagnosis Threshold:** N/A (Normative healthy screening vitals/labs) **Medical Methodology:** Pharmacokinetic Drug-Drug Interaction (DDI) Profiling **Optimization Target:** Identifying necessary label warnings or dosage adjustments for concomitant medications (metformin, digoxin, rosuvastatin, bupropion, flurbiprofen, omeprazole).

13. Study Procedures & Methodology

Management Pathway: Clinical Pharmacology Unit inpatient admission **Education Protocol (HCP):** Strict bioanalytical sampling schedule training **Education Protocol (Patient):** In-clinic restriction briefings (fasting, prohibited foods/substances) **Quality Audit Mechanism:** Real-time PK sample tracking and safety oversight

14. Observation & Follow-Up Schedule

Total Duration: Short-term (spanning ~October 2025 to March 2026 for the entire cohort) **Onsite Visit Cadence:** Intensive inpatient periods corresponding with dosing intervals **Remote Monitoring Frequency:** N/A (Conducted in-clinic) **Checklist Review Interval:** Daily during confinement periods

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |